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Innovative Approach to Assure Upper and Lower Drain Accessibility in UAE First Intervention Operation for Multilateral Level 5 Well

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Abstract

The Case study Field "A" which is one of the largest offshore oil fields in UAE, managed by the Abu Dhabi National Oil Company (ADNOC). As a mature field with complex carbonate formations, it faces ongoing production challenges due to reservoir heterogeneity.

Historically, the field has employed a dual completion design to optimize production from two distinct reservoirs. This approach, enabling simultaneous production from both upper and lower drain reservoirs within a single wellbore, effectively addresses the challenge of limited drilling slots in offshore field. However, the existing dual completion design restricts wellbore accessibility to the upper drain, limiting stimulation options to acid bullheading only. This constraint prevents the execution of coiled tubing acid stimulation in the upper drain. In contrast, the Multilateral Tie-back system (MLTBS) enables the string to extend into the upper drain's lower completion, allowing for full accessibility (mandated by Shareholders) for segmented coiled tubing acid treatments, well intervention and conformance job.

This paper presents the first intervention operation targeting both the Upper and Lower Drains through a Multilateral Tie-Back System (MLTBS) featuring a Level 5 open-hole uncemented junction. The system was specifically engineered to overcome previous accessibility limitations by enabling direct intervention into the upper drain.

By adopting the Multilateral Tie-Back System (MLTBS) approach, a transformative shift is introduced in field development strategies. As a result, it sets a new benchmark for maximizing hydrocarbon recovery from complex and mature field, offering greater operational flexibility and improved reservoir management.

Introduction

Over the course of many years, conventional multilateral well intervention operations have been constrained due to the lack of completion systems capable of addressing work-over and logging tools in the Upper drain. The industry has recognized this limitation as a significant operational challenge that impacts reservoir management effectiveness in mature fields.

In recent years, there has been a noticeable surge in conventional Multilateral Level 2 wells utilizing dual-strings completion systems for both injector and producer wells (Fig 1). This standard configuration creates a fundamental limitation: operators cannot access the upper drain while the completion is installed in the well. The lack of interventions in the upper drain would result in loss of independent surveillance and diagnostics.

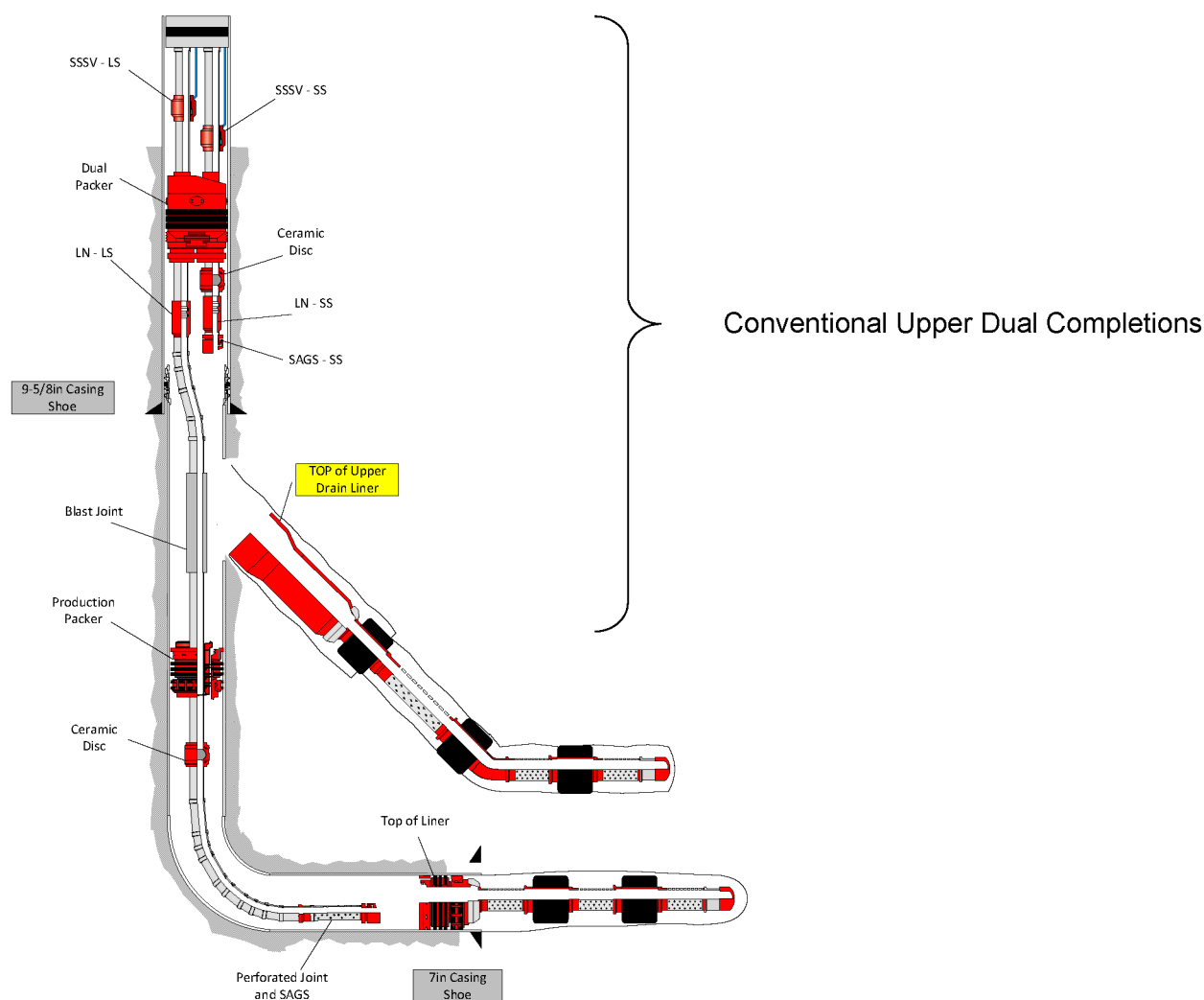


Figure 1—Schematic of the Complete conventional Dual Lateral Well

Industry demands to eliminate these complex re-entries drove a significant modification to the Multilateral system, particularly an upgrade to accommodate dual upper completion strings that tie back to the lower and upper drains lower completion. This enhancement provides direct access for well intervention to both Upper and Lower drains independently, representing a key technological improvement over conventional design.

In oil field "A", the modified level 5 MLTBS is implemented using 9-5/8" production casing to access two laterals: upper and lower drains. This casing supports the deployment of standard MLT drilling tools while enabling the construction of two 8-1/2 inch × 6 inch horizontal sections positioned optimally to achieve production zone targets successfully. The horizontal drains were completed with LEL (Limited Entry Liner) and open-hole swell packers to make the acid stimulation more effective along the entire lateral for both upper and lower drain. The LEL acts as a mechanical diverter, by allowing the distribution of the acid along

the targeted intervals in both drains. This is particularly important in a heterogeneous reservoir with varying parameters.

Current conventional system Configuration and Performance Insights

First limitation: Conventional Oil Producer and Water Injector wells don't provide access to the upper lateral, the limitation of surveillance in the upper drain creates a substantial production challenge in complex carbonate formations. When the upper lateral loses productivity due to various factors like skin damage, reservoir pressure depletion and/or water break-through, operators were forced to limited options to revive the short string production without any production monitoring / saturation tools to identify the root cause of well shut in. (Fig 1)

Second limitation: In conventional dual oil producers and injectors systems, the stimulation efficiency is often hindered by design limitations. Specifically, these setups typically leave a gap or interval between the short upper completion string and the top of the liner in the upper drain. This unconnected space can result in reduction of fluid placement, reduction in stimulation coverage, limited control during bullheading and lack of selected coiled tubing stimulation operation.

On the other hand, the Multi-Lateral Tubing-Based System (MLTBS) offers a more integrated and efficient configuration. In MLTBS, the system accommodates dual completion strings that independently tie back to the main bore lower completions in the Lower Drain and to the lateral lower completions in the Upper Drain. This symmetric string configuration eliminates the coverage gap inherent in the upper drain. This direct connection ensures:

- Improved fluid access to the target zones.
- Enhanced stimulation precision across multiple laterals.
- Minimized fluid loss or diversion.
- Provides access to the upper drain (well intervention).
- Greater operational reliability during stimulation.

By addressing the structural inefficiencies of conventional completions, MLTBS enables more effective stimulation outcomes and better reservoir contact, making it a superior choice for complex well architectures.

To address the challenges in well accessibility for short string, a Multilateral Tie-Back System (MLTBS) technology Level 5 was implemented for its capability to provide independent dual-string production to both drains and pressure-isolated junctions across the casing exit zone. These advantages help enhance overall reservoir management by tailoring the Upper completions to both reservoir zones, Upper and Lower.

Multilateral Tieback System Completion design requirements:

The Multilateral well must meet three major requirements:

1. Flow segregation between the Lower and Upper production intervals
2. Pressure isolation at the junction
3. Full tubing accessibility for intervention to both laterals.

The core of the MLTBS that was selected to meet these objectives has been designed to divert and tieback a 3-1/2" Dual Tubing completion Long and Short strings towards the relative seal-bores that are strategically placed in the Upper and Lower drains. Once the stingers are sealed into the seal-bores, the junction is qualified as a Level 5 pressure-isolated MLT system

Proposed MLTBS well design KPI

Table 1—Well construction KPI

Metrics	Proposed MLTBS system	Conventional
Full-bore access to Upper and Lower drains		
Segregated flow		
Pressure isolation at the junction		

Proposed Multilateral well technology impact on the case study offshore field

Multilateral technology is designed to enhance overall production efficiency, streamline field development processes, and support advanced data analytics. By utilizing multiple branches from a single drilling location, these wells can optimize resource extraction and improve access to different reservoirs. Given the offshore nature and the limited well slots, MLTBS technology provides a great value and low footprint without compromising any aspect that the conventional dual completions usually impose. This flexibility not only boosts production rates but also allows for monitoring and management of reservoir performance, ultimately leading to qualified decision-making based on real-time data insights.

As a result, multilateral technology plays a crucial role in driving innovation and efficiency across the oil and gas industry.

Oil producer MLTBS wells

These wells are designed for production, with laterals placed in the same or different reservoirs.

The multilateral access window offers the flexibility to access each lateral at any stage of the well's lifecycle, enabling efficient data acquisition and optimized intervention planning. Additionally, during an intervention targeting the upper drain, well longevity can be enhanced by deploying mechanical intervention, such as coiled tubing equipped with a shifting tool to shift the ICD sleeve. This approach helps to delay water encroachment and sustain effective production performance.

Water Injector MLTBS wells

These wells are designed for injection operations, with both laterals placed in either the same or different reservoirs. The use of multilateral technology enhances overall well integrity and, when integrated with a multilateral access window, facilitates targeted interventions to optimize injection performance and data acquisition. Moreover, during an intervention in the upper drain, mechanical access such as coiled tubing equipped with a shifting tool can be utilized to shift the ICD sleeve for the water injector in high perm streak interval to open and /or closed position. This proactive measure helps delay water breakthroughs in adjacent producer wells, thereby extending the effective life of the oil producer wells.

MLTBS Well construction and optimized operations

Run Lateral Completion and Retrieve Whipstock (one trip)

The first part of the future well envelope was positioning the lower completion in the Upper drain. Once the window in a motherbore casing is milled, the lower completion was run and strategically placed in the UD, with the positioning of the lower completions and zonal isolation packer in the confirmed zones. (Fig 2)

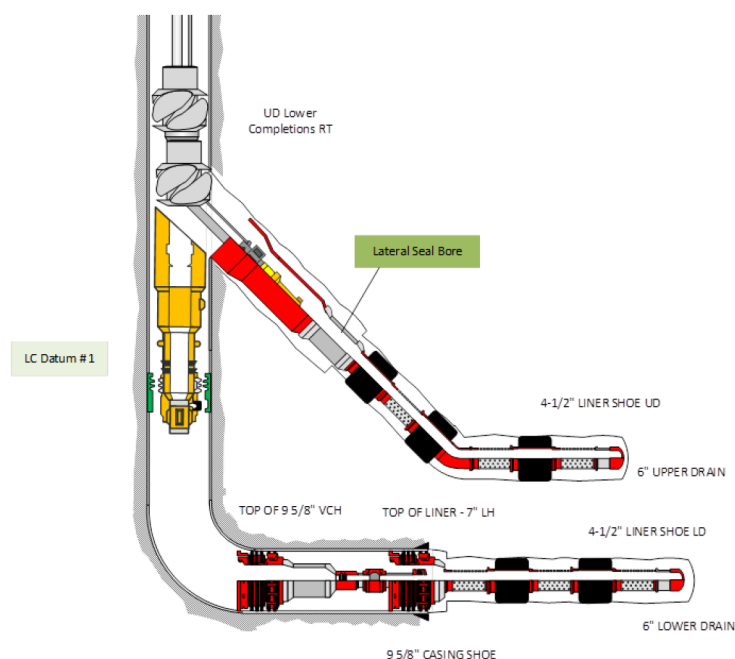


Figure 2—Upper Drain Lower completions positioning Run Mainbore Intermediate Completion

The proposed system was successfully executed, the Lower completions system was positioned in the Upper Drain within 10 ft from the Bottom of the Window in 9-5/8in Casing. This precise positioning enabled optimal spacing of isolation packers within the productive interval zones, ensuring effective zonal isolation and production optimization.

The second part of the well envelope is to run and set the Intermediate mainbore completions in the casing exit interval. The assembly is providing the sealing surface that would allow for sealing of the upper completions Seal Stinger and to deflect the lateral Seal Stinger towards the Upper Drain Lower Completions Seal bore.

Once the system is run and set the Pressure test was successfully performed to confirm the Well's envelope integrity. (Fig. 3)

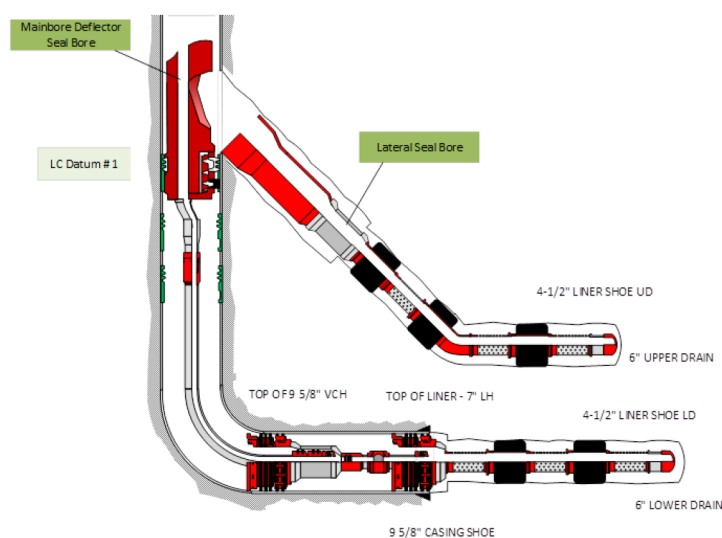


Figure 3—Schematic of the Mainbore Intermediate completions.

The **third part** of the well envelope is to run the Dual 3-1/2" Upper Completions system that would tie each leg Short String and Long String, into the Intermediate completions and the Upper Drain Lower completions, respectively. The strings and the Seal Assemblies were positioned as per the Tubing Movement modelling that would allow for the successful production of the fluid without losing the integrity of the strings. Which confirms the MLTBS system Level – as a TAML 5. Refer to (Fig.4)

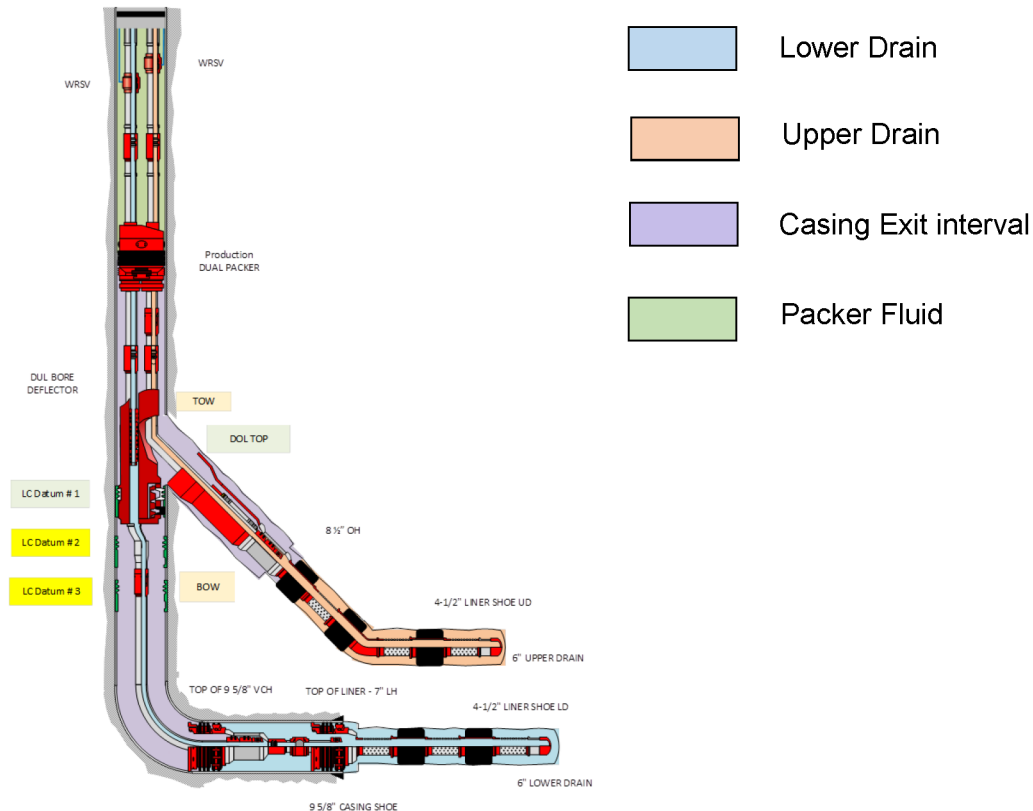


Figure 4—Final Level 5 Multilateral well configuration envelope.

Additional System benefit enabled to run the Lower Completion into the Upper Drain interval and retrieving the Whipstock at the same time.

Following successful lateral liner placement, the liner running tool (LRT) was disengaged from the liner string and raised above the whipstock. The LRT was then precisely positioned within the whipstock bore to initiate the release mechanism. The whipstock was subsequently released, operating within its designated parameters, and was retrieved to the surface.

For the first time in the Middle East, an advanced liner deployment system incorporating a newly designed running tool was successfully implemented (Fig 5).

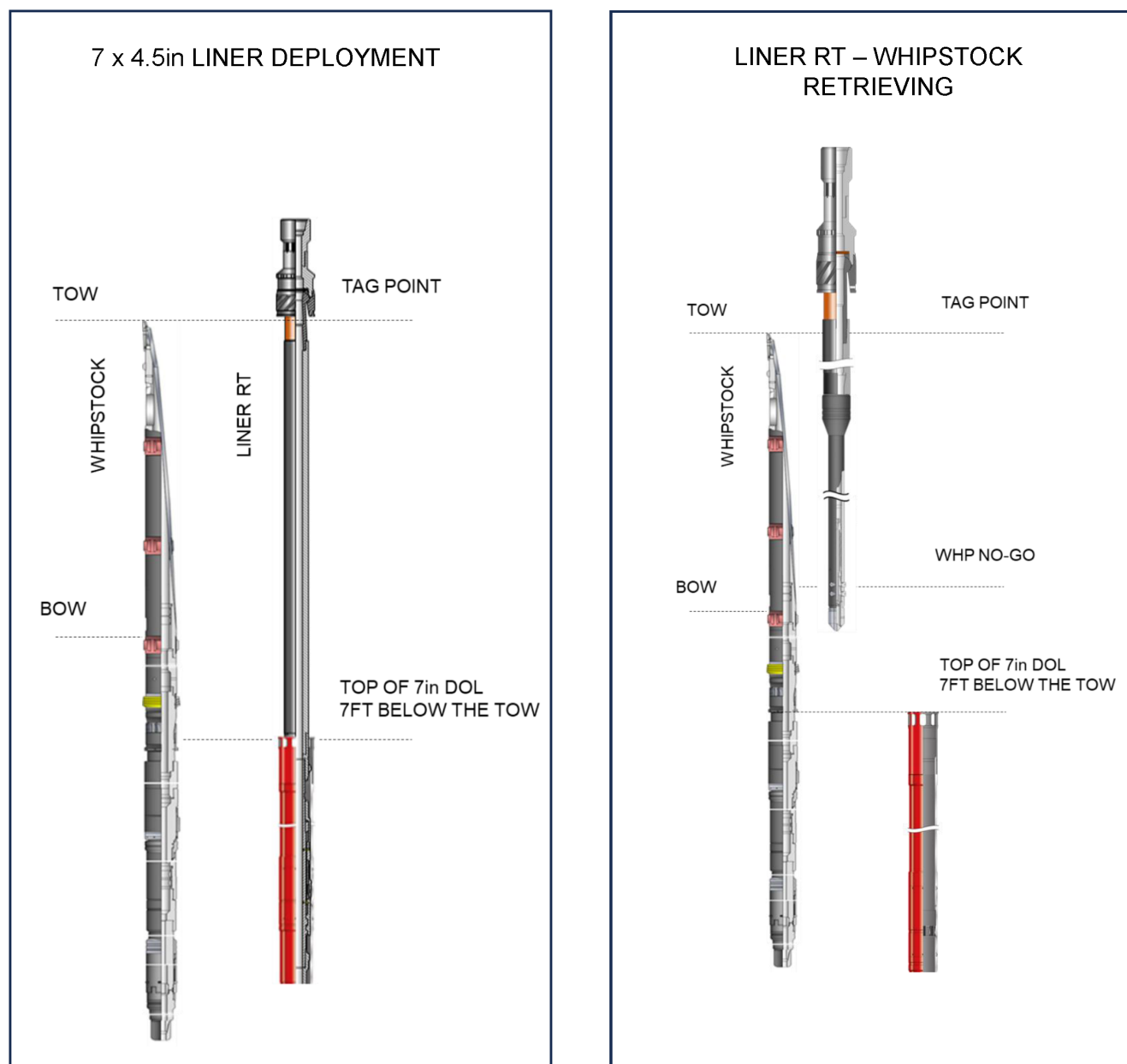


Figure 5—illustrates the simultaneous deployment of lower completions and retrieval of the whipstock within a single operational trip.

Comparison between Open-hole level 5 junction and Cemented level 5 junction

While MLTBS is conventionally deployed in dual lateral wells featuring cemented junctions with intermediate 7" cemented liners in the upper lateral, this project represented a deviation from standard practice. For the first time, the operator implemented the Level 5 system without cementing the intermediate 7" liner. Instead, an 8½" × 6" open-hole section was created in the upper drain. The 7" × 4½" lower completion assembly was run and positioned directly in the open hole, and connected via a 3½" tubing string. This tubing string was equipped with seal assemblies in a dual upper completion configuration. This system enables the construction of a pressure-isolated junction with confirmed pressure integrity. [Fig 6.](#)

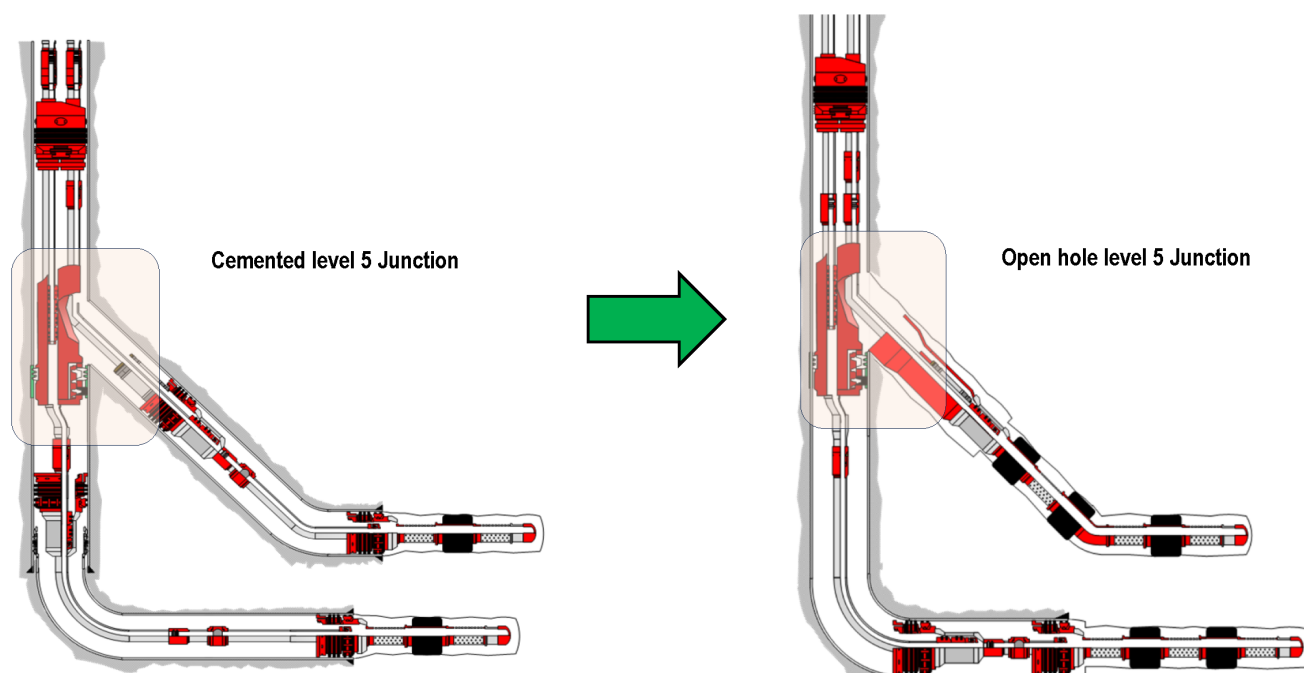


Figure 6—Comparison between cemented and open-hole Level 5 junction

Modified Level 5 systems

- Open hole junctions in multilateral completions are ideal for horizontal wells in stable, consolidated formations where borehole integrity can be maintained without casing or cement. These junctions can only be designed in cases where the target zone is just below the final casing avoiding contacting multiple zones having different pressure regime to avoid differential sticking. Moreover, they are particularly suited for zones with low pressure differentials and where zonal isolation is not critical. Compared to cemented junctions, open hole junctions offer advantages such as simpler installation, lower cost, greater reservoir contact, and operational flexibility.
- Liner hanger and isolation are utilized for the lower drain only.

Cemented level 5 systems

- Liner hanger and the isolation packers are used in both laterals, Lower and Upper drain

Benefits of the Open hole level 5 System

- 10-15 days reduction in drilling and cementing operations for the 8½" Upper Drain interval.
- Eliminates liner hanger and isolation top packer requirements in the Upper Drain.
- Simplifies system installation and avoiding operational complexity.

Results And Outcome

The deployment of the modified Level 5 MLTBS has established a new benchmark in intervention capability and reservoir management for dual lateral wells. One of the most significant milestones in this project was the successful deployment of a Production Logging Tool (PLT) into both drains as shown in (Fig 7,8,9 and 10) it was as an achievement that was previously unattainable using conventional completion systems.

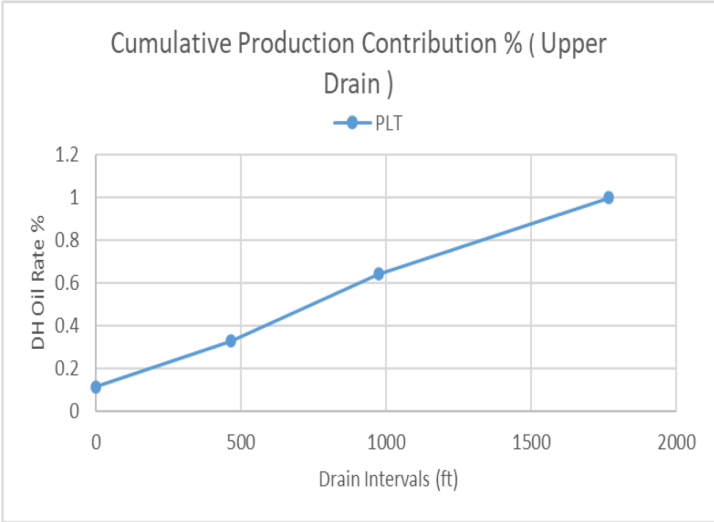


Figure 7—Cumulative Production Contribution % (Upper Drain).

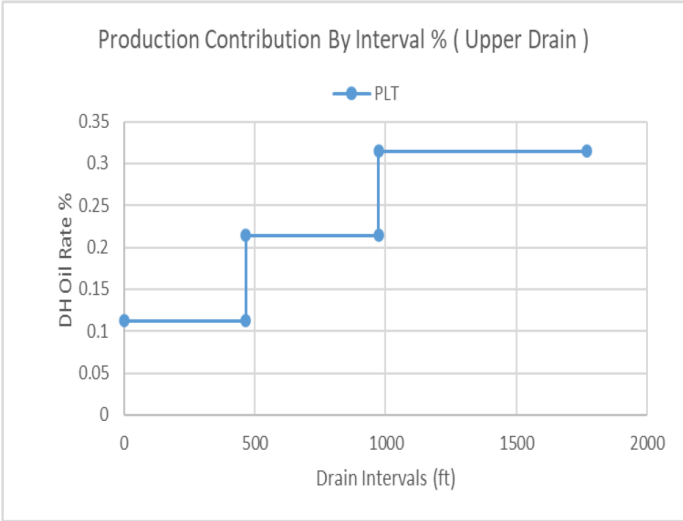


Figure 8—Production Contribution by Interval % (Upper Drain).

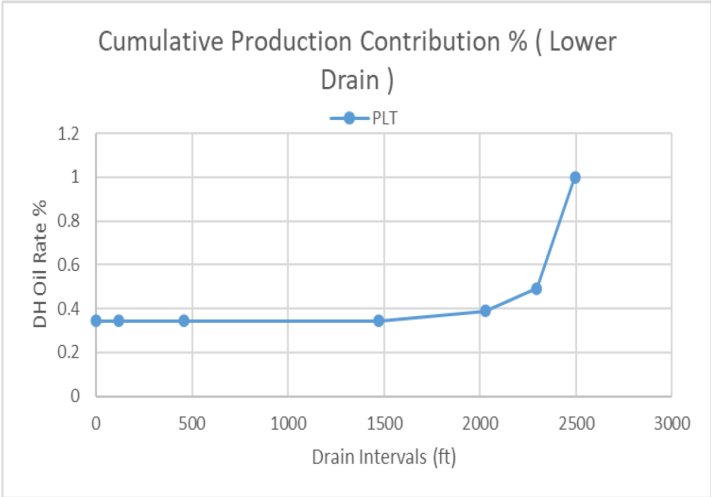


Figure 9—Cumulative Production Contribution % (Lower Drain).

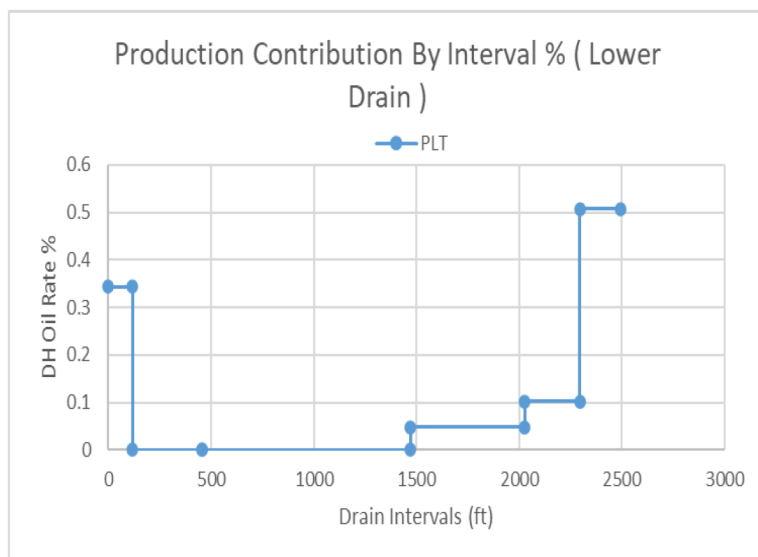


Figure 10—Production Contribution by Interval % (Lower Drain).

Historically, dual completion wells have posed major challenges for well intervention, particularly in accessing upper drains due to limited entry paths. This limitation often prevented routine surveillance, forcing reliance on surface data or modeling, which are inherently less accurate when evaluating zonal contributions and diagnosing production anomalies.

With the introduction of MLTBS, full-bore access to both laterals was achieved, enabling smooth, wireline-based interventions without requiring a workover operation. This feature provides critical operational advantages, allowing real-time monitoring and proactive maintenance of the wellbore without the risks and costs associated with intrusive interventions. The successful PLT deployment in the upper lateral not only demonstrated the system's operational reliability but also confirmed that both laterals can now be accessed with the same flexibility as a single horizontal wellbore.

One of the primary benefits of this capability is the ability to perform interventions aimed at optimizing production efficiency and addressing issues such as scaling, debris buildup, or localized flow blockages. In the past, when upper drains in conventional systems underperformed or encountered issues, operators often had to conduct expensive and time-consuming workovers, leading to production downtime. The MLTBS has eliminated this bottleneck by providing a permanent, open pathway that allows tools to be run in and out of both laterals as needed, saving both time and cost. In this case, the ability to run a PLT in the upper drain for the first time not only validated the concept but also confirmed that critical diagnostic and maintenance operations can now be carried out effectively.

In addition to improving operational flexibility, the MLTBS design offers long-term reservoir management benefits. The upper and lower drains can now be individually monitored and treated, ensuring that both zones contribute optimally to overall production. This capability is vital for mature offshore fields, where maintaining productivity often requires frequent surveillance and targeted intervention. The successful PLT deployment demonstrated that advanced surveillance techniques can now be performed routinely, enabling early identification of underperforming zones and facilitating corrective measures—such as acid stimulation or selective water shutoff—in a much more efficient and cost-effective manner.

The mechanical robustness of the MLTBS junction was also proven during the PLT intervention. The tool string passed smoothly through the junction multiple times, confirming that the system's design supports safe, reliable access to each drain without the risk of obstruction or mechanical damage. This reliability is particularly important for long-term well management, as it ensures that future interventions—whether diagnostic or remedial—can be conducted without operational constraints.

The success of the PLT intervention reinforced the importance of integrating advanced completion systems with proactive surveillance strategies. By enabling easy re-entry and tool conveyance, the MLTBS creates a framework for continuous reservoir monitoring. Stakeholders now have confidence that critical diagnostics can be performed at any stage of the well's life cycle, ensuring that production strategies can be adjusted in real time to reflect changing reservoir conditions. This capability represents a fundamental shift from reactive to proactive well management, ultimately extending the productive life of the well while optimizing overall recovery.

The operational outcomes of this intervention extend beyond a single well. By successfully running a PLT in the upper drain, the project has proven the scalability of MLTBS technology to other wells within the field. Many offshore wells face similar challenges, with multiple laterals that are difficult to monitor and optimize. The deployment of MLTBS provides a standardized solution for such scenarios, enabling operators to achieve higher recovery factors while reducing operational costs. The system's design is particularly advantageous in carbonate reservoirs, where heterogeneity and varying permeability often demand a high level of surveillance and zonal control to optimize production.

Conclusion

The deployment of the MLTBS in an offshore environment with limited slots to drill single completion wells represents a major advancement in well intervention and reservoir management for dual-lateral completions. This innovative system introduces full-bore access to both the upper and lower drains, effectively resolving a long-standing operational challenge inherent in conventional completion designs. By removing access limitations, MLTBS empowers operators to conduct targeted surveillance, diagnostics, and remedial operations in each lateral independently—without the need for disruptive and costly workover procedures.

Traditionally, restricted access to upper drains has forced reliance on indirect surveillance techniques such as surface data interpretation and reservoir modeling. These methods, while useful, often introduce uncertainty in evaluating zonal performance and diagnosing production anomalies. The MLTBS design eliminates these constraints by providing direct, reliable access to both laterals. This enables accurate performance monitoring, timely optimization of production strategies, and more informed reservoir management decisions. Such precision is particularly valuable in heterogeneous carbonate formations, where production variability between laterals can be substantial and difficult to manage without direct intervention.

Operational results from this deployment confirm that advanced multilateral completion technologies like MLTBS can be reliably implemented in complex offshore environments. The system's robust, pressure-isolated junction maintained full integrity throughout multiple intervention runs, demonstrating its mechanical resilience and suitability for repeated use over the well's lifecycle. Importantly, the ability to perform interventions while maintaining production from both laterals significantly reduces downtime, minimizes deferred production, and enhances the overall economic performance of the asset.

From a reservoir management standpoint, the capability to isolate, monitor, and treat each drain individually allows for tailored production optimization. Localized issues—such as scaling, water breakthrough, or debris accumulation—can now be addressed precisely, without affecting unaffected zones. This selective intervention capability not only extends the productive life of the well but also supports sustained production rates and improves overall recovery potential.

Beyond its immediate operational benefits, the successful deployment of MLTBS validates its scalability as a standardized solution for similar wells within the same field or across other offshore developments. Standardizing this approach streamlines planning, reduces execution time and cost, and establishes a repeatable methodology for managing dual-lateral wells. This consistency is especially beneficial for

field-wide implementation, where uniformity in design and execution can lead to significant operational efficiencies.

In summary, the modified Level 5 MLTBS has proven its ability to deliver measurable improvements in offshore well intervention efficiency, operational flexibility, and reservoir management. By overcoming the access limitations of conventional dual completions and enabling proactive, precise interventions, the system provides operators with a practical and repeatable method for optimizing production performance. Its demonstrated reliability, scalability, and adaptability to complex carbonate reservoirs position it as a best-practice solution for modern offshore field development and long-term well management.

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